Code No: 117DX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, June/July - 2024
INFORMATION RETRIEVAL SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used Papers 2024

PART – A

(25 Marks)

1.a)	Define information retrieval.	[2]
b)	What is a term frequency and normalized term frequency?	[3]
c)	What is computing New Query Weights?	[2]
d)	What is Ward's method in clustering? Give example.	[3]
e)	Define index pruning.	[2]
f)	Explain about Pharse Translation.	[3]
g)	Define fusion.	[2]
h)	Describe compression based on posting list size.	[3]
i)	Describe about Query Log Analysis.	[2]
j)	What is high-precision search?	[3]

PART – B

(50 Marks)

2.a)	Explain about retrieval strategies and their categories.	
b)	What is smoothing in language model? Explain.	[5+5]
	OR	
3.	Explain key concerns with Probabilistic Retrieval Strategies in detail.	[10]
4.	Consider the following example and find the Term co-occurrence D1 : "a dog will bark at a cat in a tree" D2 : "ants eat the bark of a tree".	[10]
	OR	
5.a)	Write a short note on hierarchal agglomerative clustering.	
b)	Explain in detail how clustering technique is used for generating Thesaurus.	[5+5]
6.a)	What are the four core questions to cross the language barrier?	
b)	Explain about document translations and query translations.	[5+5]
	OR	
7.a)	Explain how parsing of single terms is performed in information retrieval.	
b)	Explain in detail about K-Distance measure.	[5+5]

8. How do you perform Duplicate Document Detection? Explain in detail. [10]

OR

9. Explain in detail the partial result set retrieval and vector space simplifications. [10]

10. How to compute the relevance using Unchanged SQL? Explain with example. [10]

OR

11. What is distributed document retrieval? Explain the theoretical model of distributed retrieval. [10]

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